

AGENTS.md — Read this first

This repo is built from one authoritative spec:

→ `MARKOS_BUILD_SPEC.md` is the complete source of truth. Read it fully before writing any code.

TL;DR

- Product: **MARKOS AI** — an AI marketing OS for small businesses in **Bahrain**. **v1.0 is Instagram-only, bilingual (Arabic + English, RTL-first)**.
- It learns a business in onboarding, stores it permanently in a **Knowledge Vault (RAG over pgvector)**, and uses **8 AI agents** to generate strategy + content, schedule & publish to Instagram, analyze, and advise.
- **Three services:** Web (Next.js 14) ↔ API gateway (Node/Fastify) ↔ AI service (Python/FastAPI). Workers (BullMQ/Redis) for publish/analytics/reports/email/renewals.
- Data: **PostgreSQL 16 + pgvector**, Redis 7, OpenSearch, S3 + CloudFront. Payments: **CrediMax/BENEFIT (Bahrain) + Stripe fallback, 10% VAT**. Hosting: **AWS me-south-1**. Social: Meta Instagram Graph API.

How to build

Execute the milestones in `MARKOS_BUILD_SPEC.md` §22 (M0 → M6), in order. Start at **M0 (Foundation)** and submit Meta App Review the same week. Within a milestone, work in vertical slices (DB → API → AI → web UI → tests). Run `make verify` after each slice and keep CI green.

Golden rules (invariants — violating any is a build failure)

1. This file and `MARKOS_BUILD_SPEC.md` win over any other assumption.
2. Build the **Foundation (§7)** before features; build the **Vault + RAG (§12)** before the **8 agents (§13)**.
3. **workspace_id isolation is sacred** — scope every query; isolation test per table (§9).
4. **Models are config, not constants** — all model calls go through the provider interface (§13.1).
5. Respect Instagram's real limits (§18): two-step container publish, public media URL, ~25-100 posts/account/24h, App Review in M0.
6. Money = integer minor units + ISO-4217 (BHD ×1000); handle 10% VAT (§19.3).
7. Meter every AI token + image; enforce plan quotas (§19.4).
8. TypeScript strict / typed Python, no `any`; bilingual + RTL from the first screen; tests as you go; Conventional Commits; keep CI green.

When the spec is silent, choose the smallest conventional option consistent with §3 and record it in `/docs/decisions.md`.